MASTER OF BUSINESS ADMINISTRATION

SEMESTER 3

DADS304

VISUALIZATION

Unit 11

Dashboard and Story Telling

Table of Contents

SL No	Topic	Fig No / Table / Graph	SAQ / Activity	Page No
1	Introduction			3
	1.1 Learning Objectives			
2	Aggregating dimensionalities	1, 2, 3, 4, 5, 6, 7, 8, 9		4-12
3	Steps to calculate aggregating dimensionalities	10, 11, 12, 13	1	13-16
4	INCLUDE Level of detailed expressions	14, 15, 16, 17	2	17-21
5	Steps to calculate level of detail expression	20		21-23
6	Exclude level of detail expression		3	24-27
7	Nested lod			28
8	Summary			28
9	Glossary			28
10	Caselet			29
11	Conceptual Map			30

1. INTRODUCTION

A dashboard is a visual display of the most critical information needed to achieve one or more business objectives which fits entirely on a single computer screen so that all information can be monitored in a single glance. A story on the other hand, contains multiple sheets or dashboards that are combined together to convey a particular business story which shows how various facts or incidents are connected and what can be done or could have been done to improve a business outcome. In this chapter, we are going to look at the retail dashboard and story that have been built using the retail data set.

1.1 Learning Objectives

After studying this unit, you will be able to:

- *Explain features of Dashboard and story*
- *Explain the steps to build dashboards*
- *Steps to build a story*

2. DASHBOARD AND STORY USING TABLEAU

This section describes about dashboard and stories using Tableau. Dashboards are a way to display a wide range of data visually. A dashboard conveys different, yet related information in an easy-to-digest format. In many cases, this includes key performance indicators (KPIs) or other important business metrics that stakeholders need to understand quickly. Due to their highly customizable nature, dashboards are useful across a wide range of industries and verticals. Data of all kinds can be included in these reports with varying date ranges to help you understand: what happened, why it happened, what may happen, and what you should do. Additionally, dashboards can be understood by those who aren't as familiar with the data since they use visuals like tables, graphs, and charts.

STEP 1: A dashboard with a comprehensive view of the sales trend for the last two years is shown, and then the sales broken down by the states and then which cities are the top performing in terms of the sales.

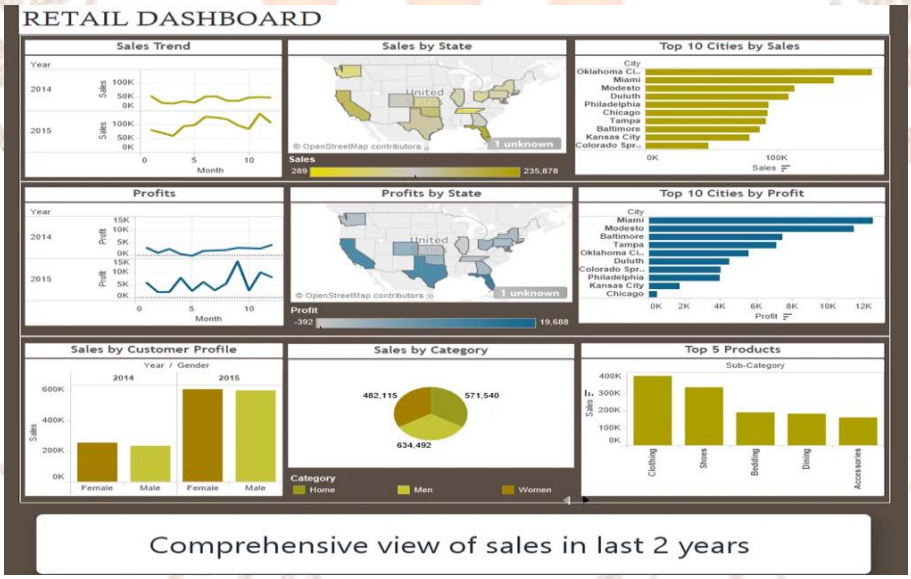


Fig 11.1 Comprehensive view of sales in last 2 years

STEP 2: The profits for 2014 and 2015 is being displayed, and then the profits is broken down by the state and the top performing cities by profit. All these views are built to be interactive as they allow us to filter to a particular month or a year or even into a particular

state. For instance, if we click on 2015 year, then all the views filter to list the data corresponding to the year 2015 alone. Now clicking outside will return us back to the original state and in the same single view we can also figure out the top five products for a particular state or even the overall sales by category and customer profile as and if more purchases are being made by male or female and so on.

Fig 11.2 Retail dashboard

STEP 3 : A dashboard gives us so much of information and the interactive capability allows us to derive insights and answers questions that might come up as we review this dashboard. And with dashboards we can also have multiple tabs representing a different dimension of

the same business information or other aspects of the business contributing to the business performance or outcome. On looking at the second tab, sales by category, we have the sales broken down by the category and the subcategory, along with the delivery status, both overall and the various location it is shipped from. This view is again interactive, allowing us to filter by the year or the month.

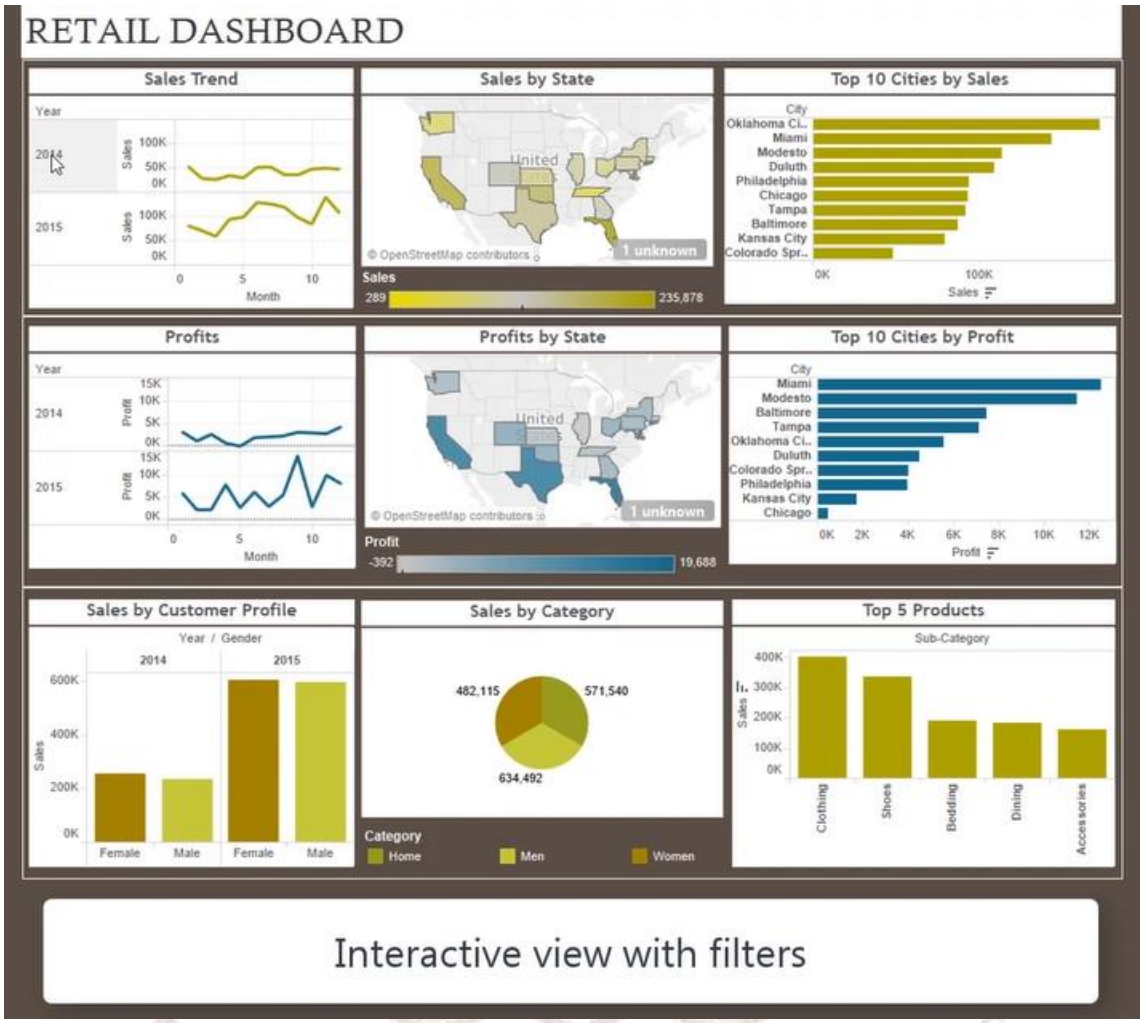


Fig 11.3 Interactive view with filters

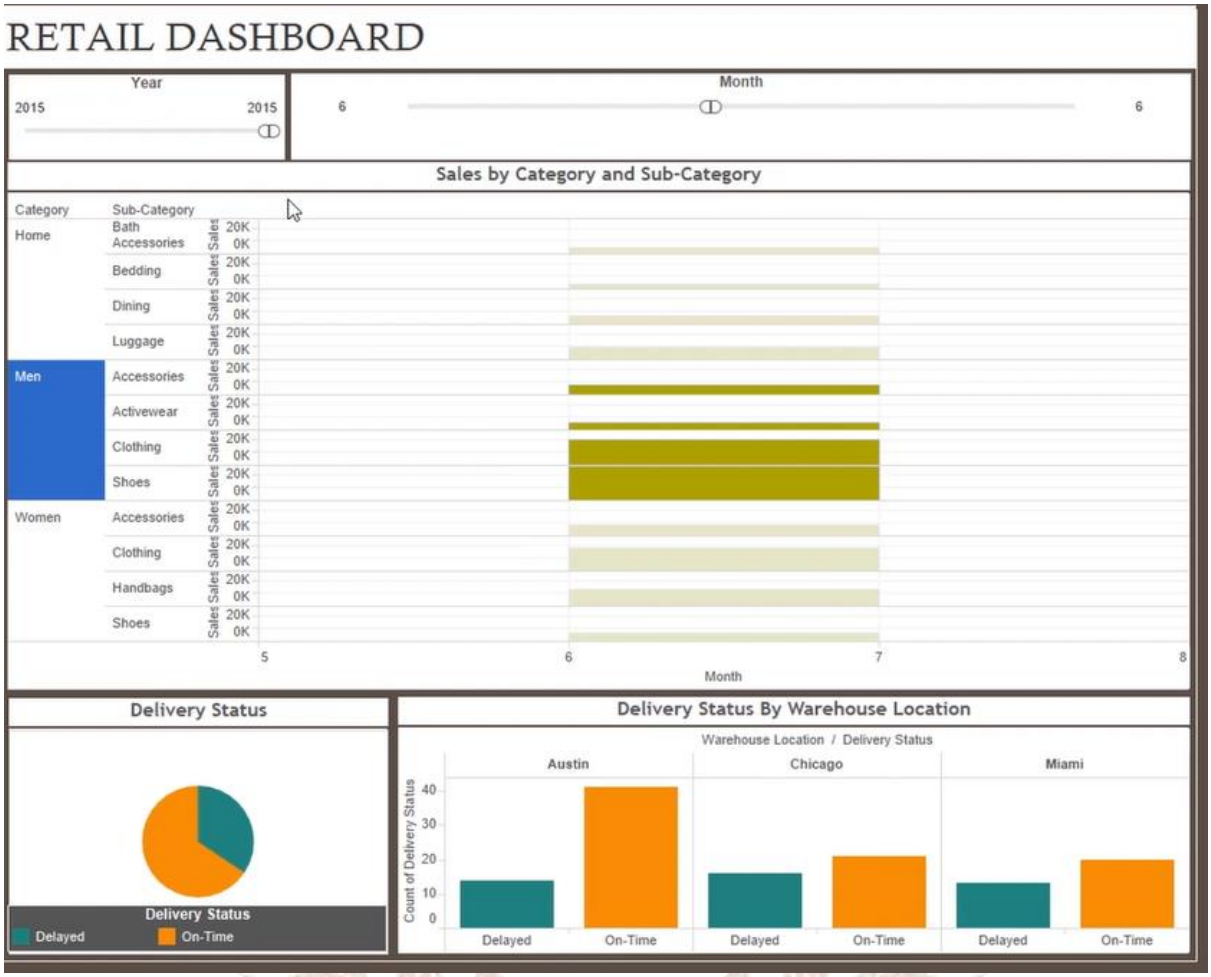
STEP 4 : And also clicking on the individual categories will allow us to monitor the delivery status for that particular category. Now by clicking on various months or the years or the categories, we can see if we have a problem with a particular location or a category and take any timely decision that's necessary to resolve any issues that exist. And lastly, we have a tab that displays the sales broken down with a customer profile, or rather gender in this case. This again gives a comprehensive view of how gender influences the sales, both regionally

and by category. So we have seen a dashboard that contains all the critical information within a single view or tabs, allowing the user to monitor his business performance effectively.

Fig 11.4 Overall sales by category and customer profile

STEP 5 : Again, each tab year that we saw is a single view of a particular business function or area that the user is interested in analyzing and deriving Insights . In the dashboard, we saw the way various visualizations helps the user analyze the data, but in a story, we're actually presenting our findings from the analysis in a conclusive manner to the end user. So different facts are knit together to form the story that conveys why a particular scenario happened and what could have been done to prevent something bad from happening or make some conclusive recommendation based on the facts here. From an analysis that has been done using the retail data set, we've seen that the sales went down in August and

September of 2015. And our story basically is targeted at presenting our findings related to this decline in sales and why it happened and what could have been done to prevent this scenario.



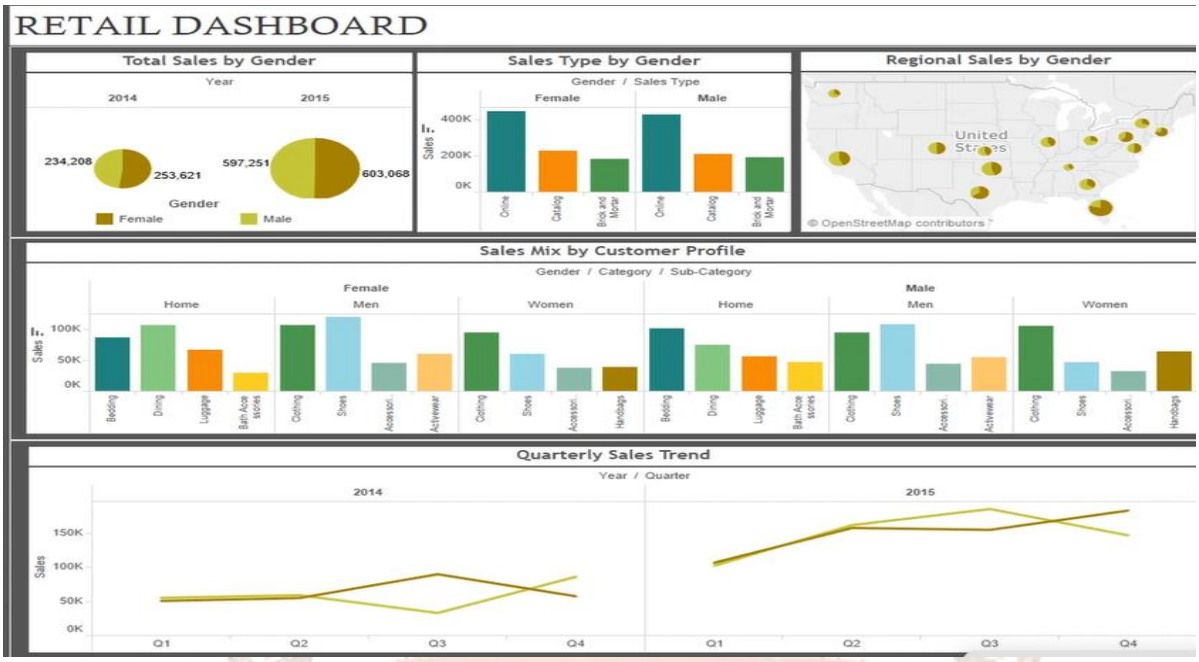


Fig 11.5 Sales by category and sub-category

STEP 6 : The story that we've built, starts with displaying the sales trend across 2014 and 2015 by women, men and home. The first tab of the view shows how sales has taken a huge dip in August and September of 2015, especially for the women category. Comparing this to the previous year, we see that this is not a seasonal dip as previous year's sales seems to be fine during the same time period and this exactly was conveyed as part of this initial story view. And then we move on to the next tab where we further go into the details as to which women's categories experienced this particular decline. Here we see that the clothing and category sales have particularly gone down significantly in both of these months.

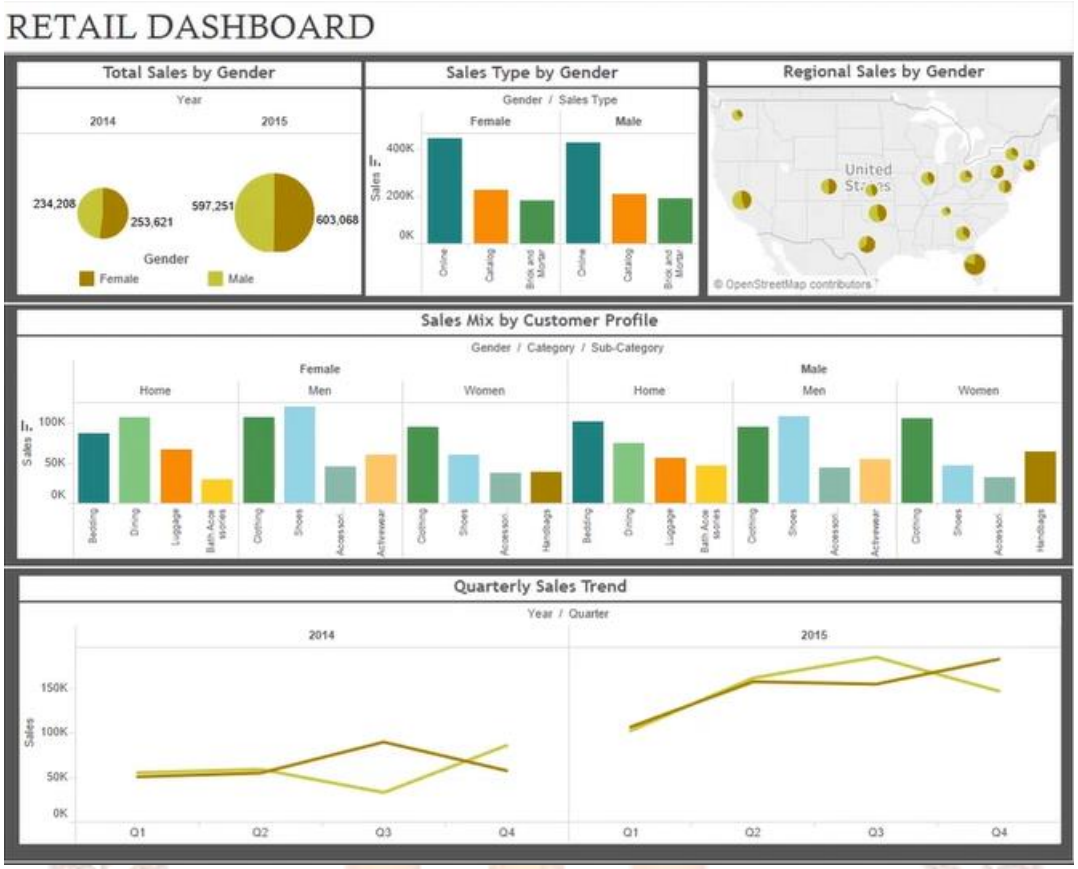


Fig 11.6 Building story

STEP 7 : Go ahead and look further at the customer sentiment for these months. Here we see that there is significant negative sentiment factor associated with June and July of 2015 and analyzing these sentiments by clicking on these months here further shows that a lot of it is due to delayed or bad service. So we move on and analyze the delivery status for these particular months. Here it is noticed that the clothing and the handbag shipments particularly were delayed during June and July due to bad weather conditions in Chicago. This in turn led to a lot of social media negative sentiment which we saw in the previous tab. So this affected the sales for the next two months as customers weren't too keen on ordering given the shipments were delayed.

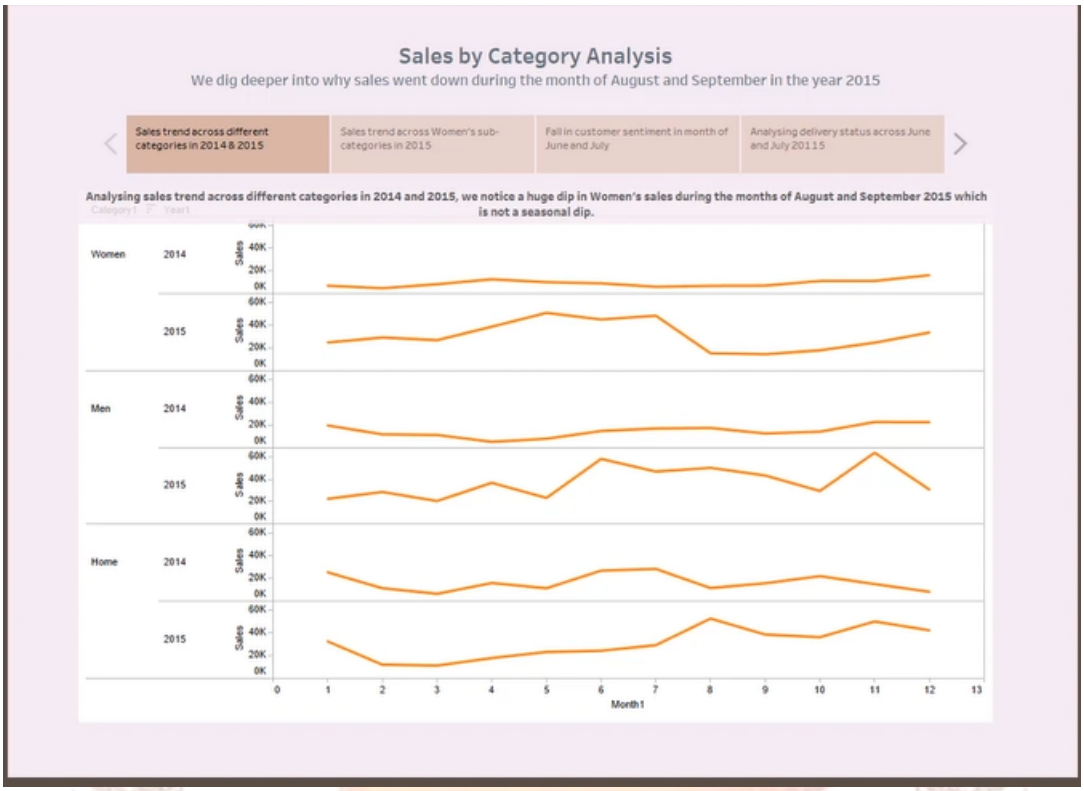


Fig 11.7 Different facts are knit together to form the story

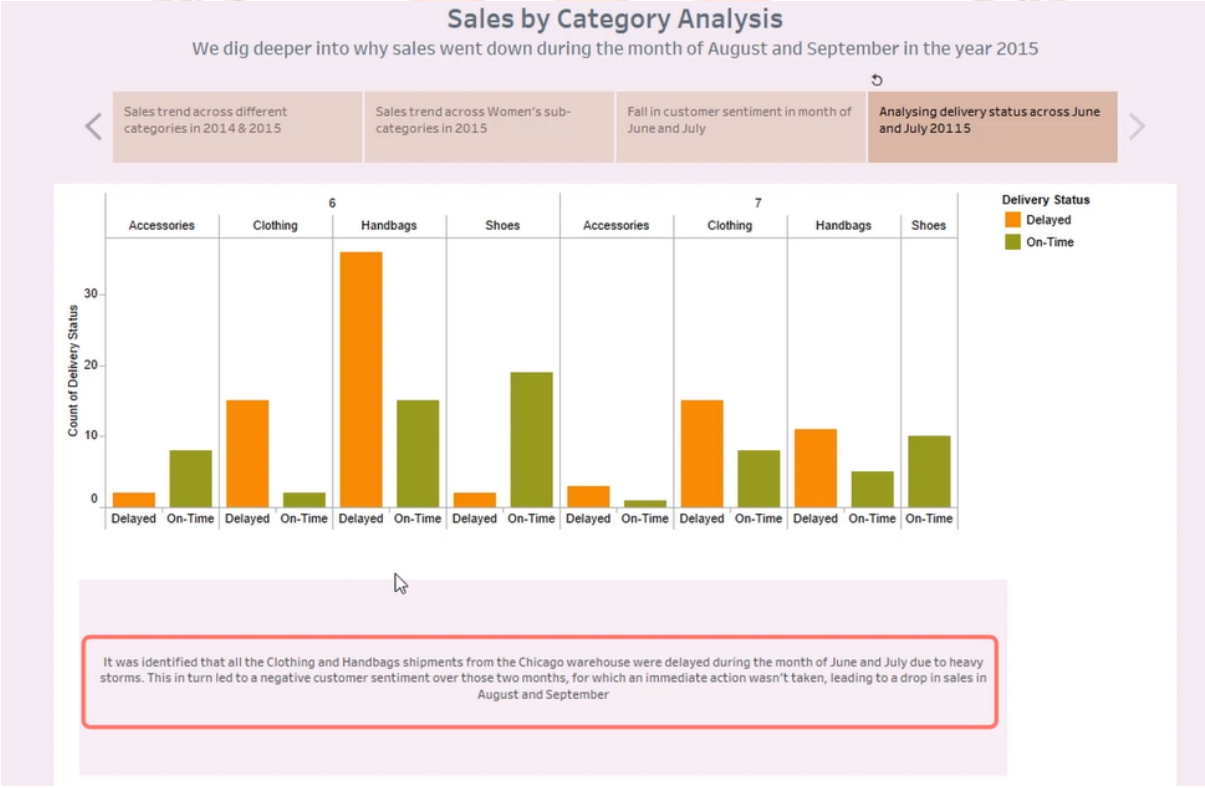


Fig 11.8 Sales by category analysis

STEP 8:

If this had been addressed sooner and the company had clarified to the customers the reasons for the delay and how this was a temporary problem and that they were working actively on it, it would have helped them avoid the negative impact on the sales. So in the story that all these facts are put together the steps that could have been adopted to improve has been illustrated.

Fig 11.9 This Story contains all facts and recommendations for an improved business outcome

3. BUILDING DASHBOARDS

Once the views have been built, the next logical step would be to use a dashboard or a story to present these data views. To create a dashboard, we can either click on the new dashboard icon next to the ad sheet icon, or click on the dashboard link on the top menu ribbon.

Step 1: Initially the dashboard workspace has opened up. A dashboard pane on the left opens up with the dashboard properties in place of the data pane. This pane displays all available sheets that can be used in the dashboard.

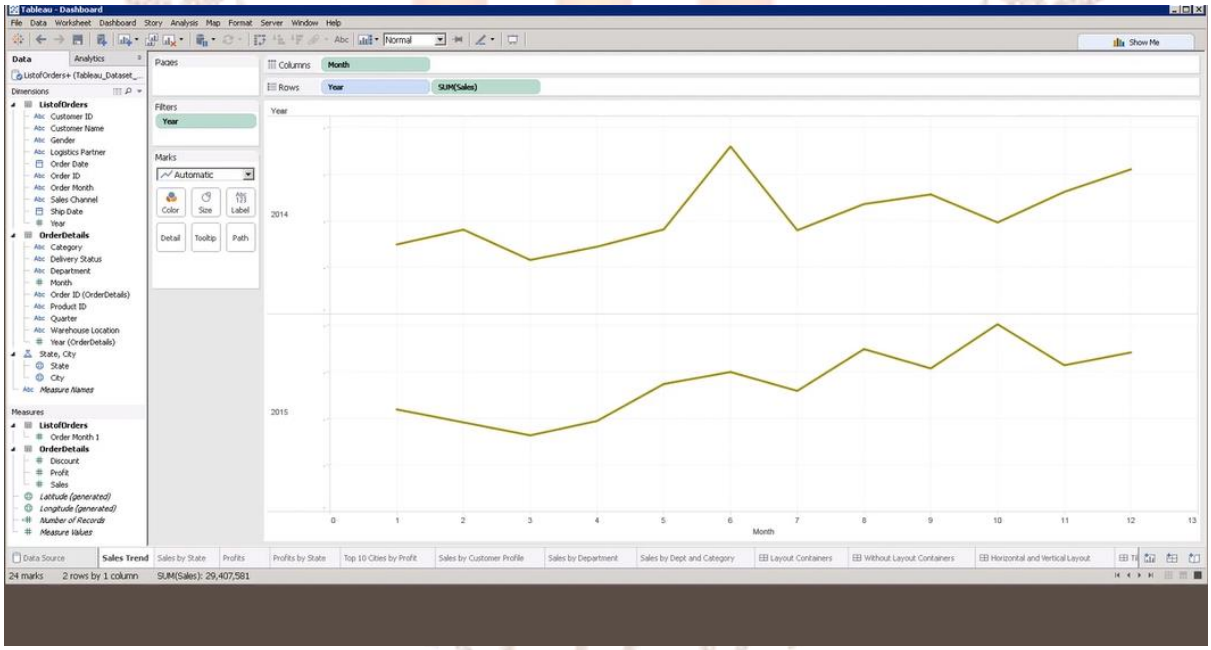


Fig 11.10 Building dashboard and story

Step 2: The first thing while building dashboard is to set the size of the dashboard in the size drop down. We have various options from automatic to exactly to range or various portrait

and landscape options and also an option for iPad. Automatic is an option that the dashboard determines to fill on its own within the available area, exactly as the name suggests gives us the flexibility to set some fixed size that we want the dashboard to display in. Then there is the select range option. This option is used to set the limits or boundaries that the dashboard can expand or shrink into. It's very important to plan ahead and set the size so that we have to end up redesigning the layout at a later point. Add the views to the dashboard. It's as easy as dragging and dropping a view to the dashboard layout. When we drag the sheet one, it takes out the entire space. Go ahead and drag the next view.

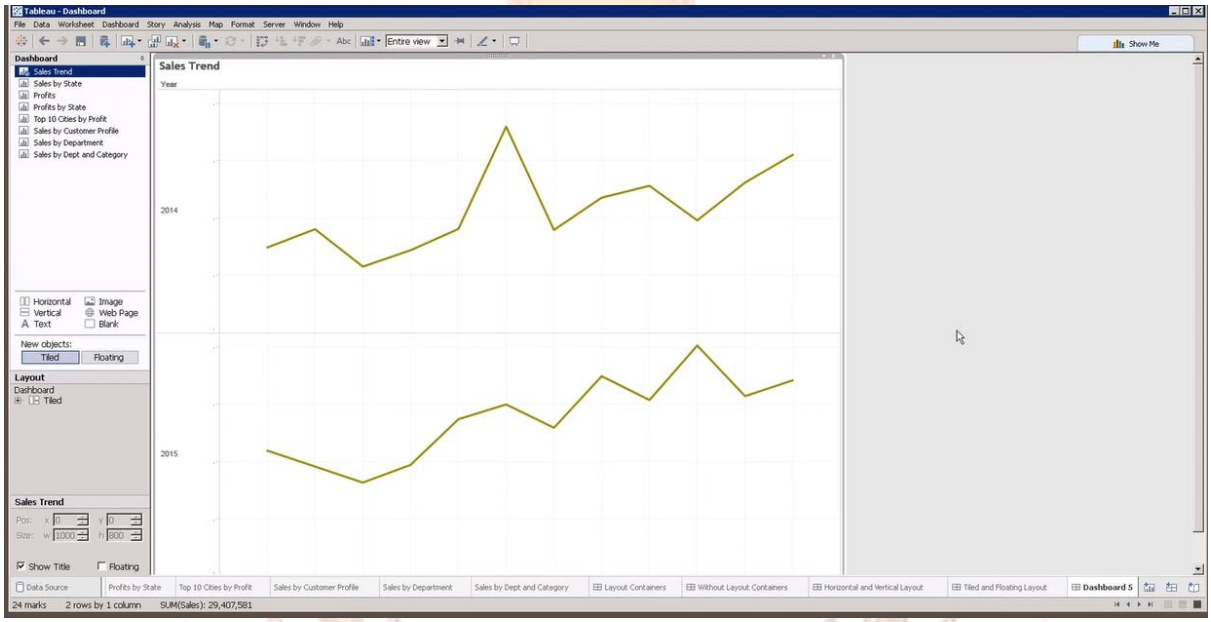


Fig 11.11 Adding views to the dashboard

Step 3: Once dragged and dropped a view, a checkmark icon appears next to the view indicating that the view is used by the dashboard. Secondly, the options on the left panel are available to format the dashboard layout. Initially, we have a horizontal container, a vertical container, images, web pages, text and a blank box that can be added to the dashboard. Before we dig deeper into the containers. Tile layouts are the ones that are arranged in a single layer grid that adjusts in size based on the total size of the dashboard objects, whereas floating objects are the ones that can be layered on top of other objects and have a fixed custom position and size. Drag the views using the tile layout option. The objects are arranged next to each other using a horizontal layout container for both the top and the bottom row. Change these views to use the floating layout.

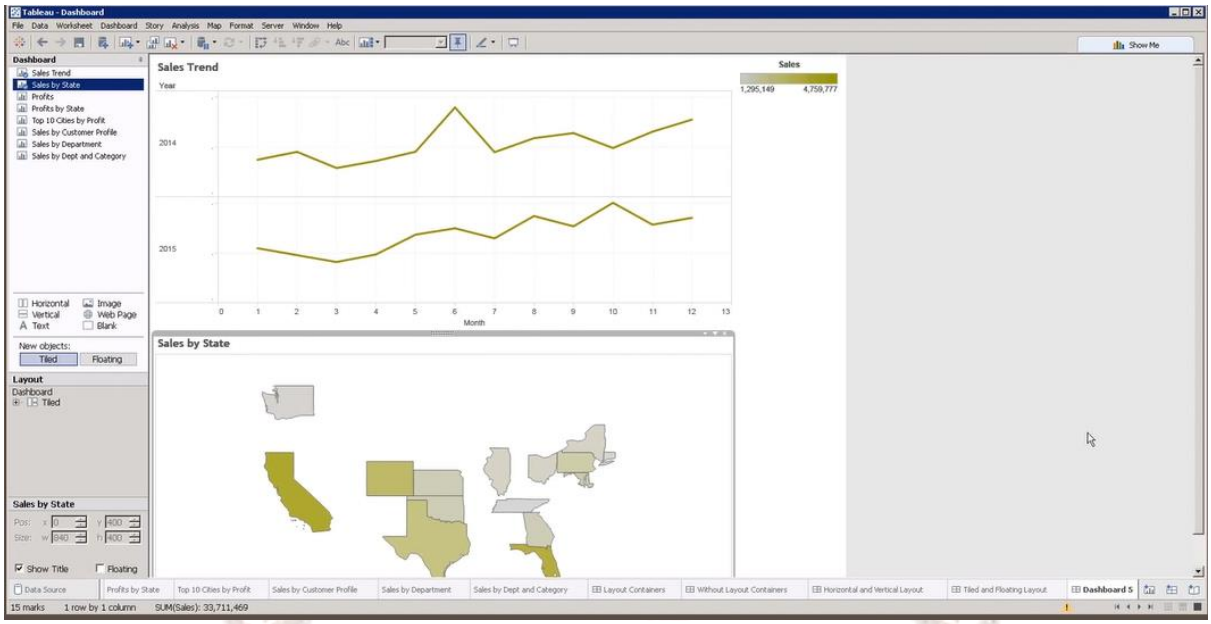


Fig 11.12 Sales trend and sales by state

Step 4: The second view is layered on top of the first view. It's also possible to switch between tiled and floating by clicking on the floating option from the shortcut menu. Change the tile layout option for the first view. So now both views are floating. In the layout section on the left all the items that have been added to the Dashboard are available. The order of these items can be changed by dragging and dropping items in a different order than the hierarchy. Tile layout items can never be reordered. Change the tile layout option for the Sales Trend view. Now change this to floating and position each one of these on top of each other. The order in which it is arranged in the layout on the left determines where both these views are placed.

Step 5: The order has been changed. Move Sales by State to the top. The order has been changed and the item Sales by Trend automatically goes to the back. The order of the items can be changed by dragging and dropping items in a different order rather than the hierarchy only for floating layout items, tile layout items can never be reordered. Next, is the position and the size options. The position allows us to specify the position in pixels where they need to be placed. And the size determines the width and height of the view for the floating objects in the Dashboard. We want the Sales by State view to be displayed or placed in the top left corner. Then specify the Y position as Zero for Sales by State.

Step 6: The Sales by state view has moved to the top left corner. The height or width for the floating object can be controlled. Then the show title option is displayed. This can be used to toggle the display for the title at both the sheet level and the dashboard level.

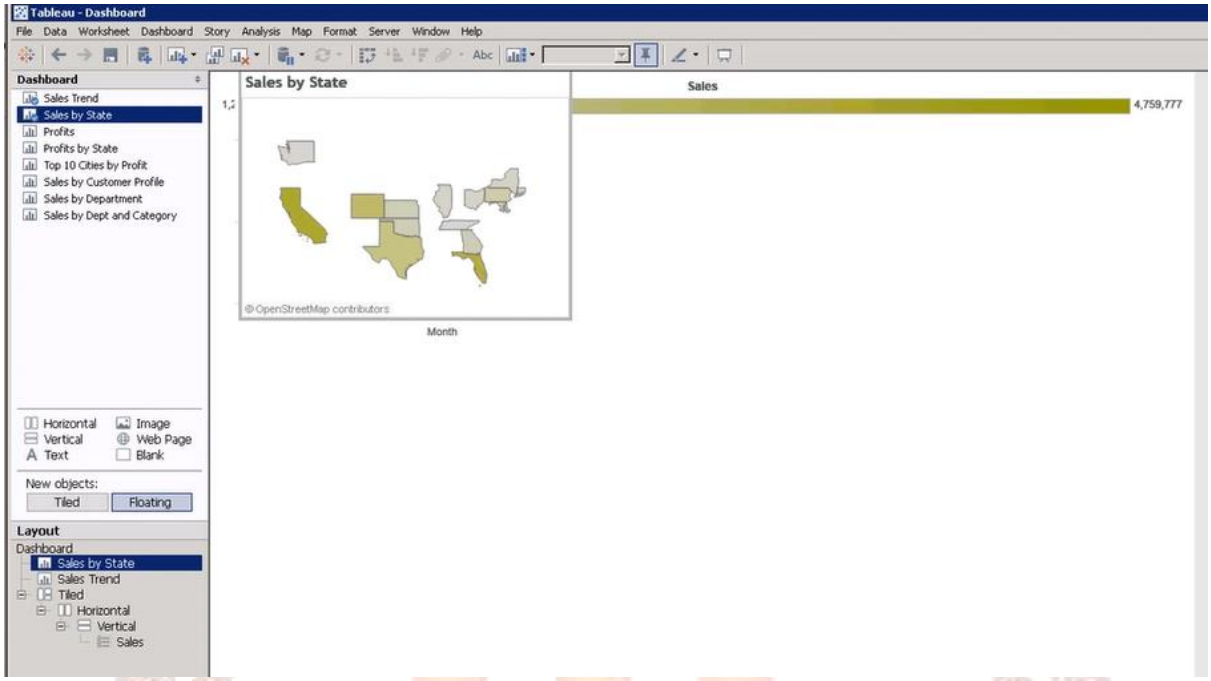


Fig 11.13 Sales by state

SELF-ASSESSMENT QUESTIONS – 1

1. _____ is an important component in a dashboard.
2. Click on _____ icon to build a dashboard.

4. FORMATTING DASHBOARDS

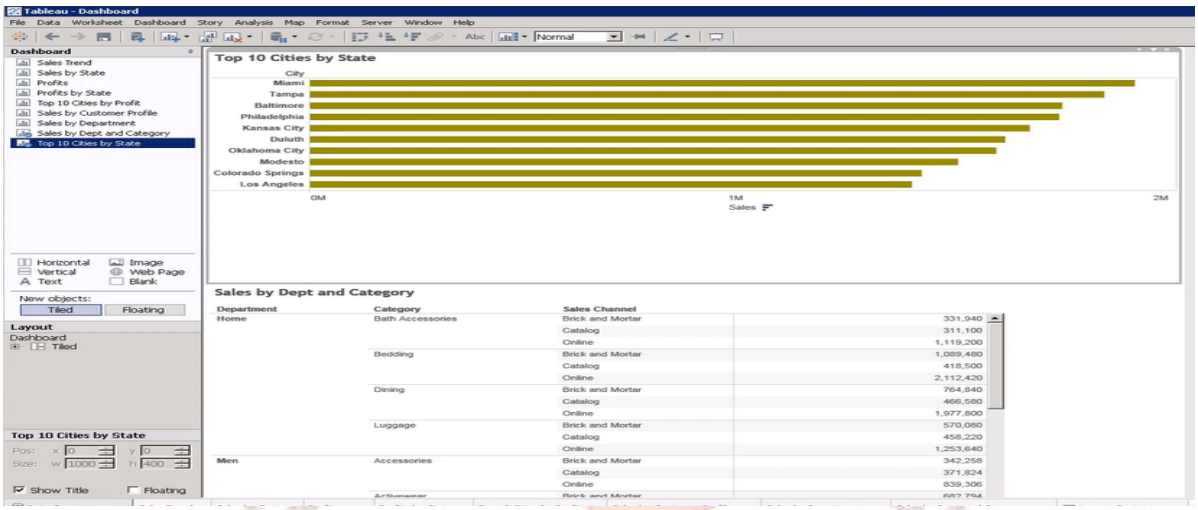


Fig 11.14 Sales by Dept and category

For formatting dashboards, consider an example which filters the top view, that is the top ten cities by state to just show Los Angeles. It is noticed that the views remain set in the place that they are placed initially, although there is enough space for the bottom view, that is the sales per category and department view to move up now and use up all the free space. A dashboard have the two views they placed inside the vertical layout container. Go in and apply the same filter here. The second view at the bottom automatically moves up using the empty space and in the process eliminating the need for a scrollbar. These container objects helps in creating the seamless experience for users by repositioning and resizing the dashboard objects whenever necessary.

Step 1: First, there is the horizontal layout container. This allows to group worksheets and any other components that needs to be part of your dashboard from left to right direction. And also using this layout, you can edit the height of all the objects that are placed within this container at the same time. And then there are the vertical containers. Vertical container allows to group worksheets and other dashboard components from top to bottom. And using a vertical layout container, it is possible to edit the width of all the objects that are placed within it at the same time.

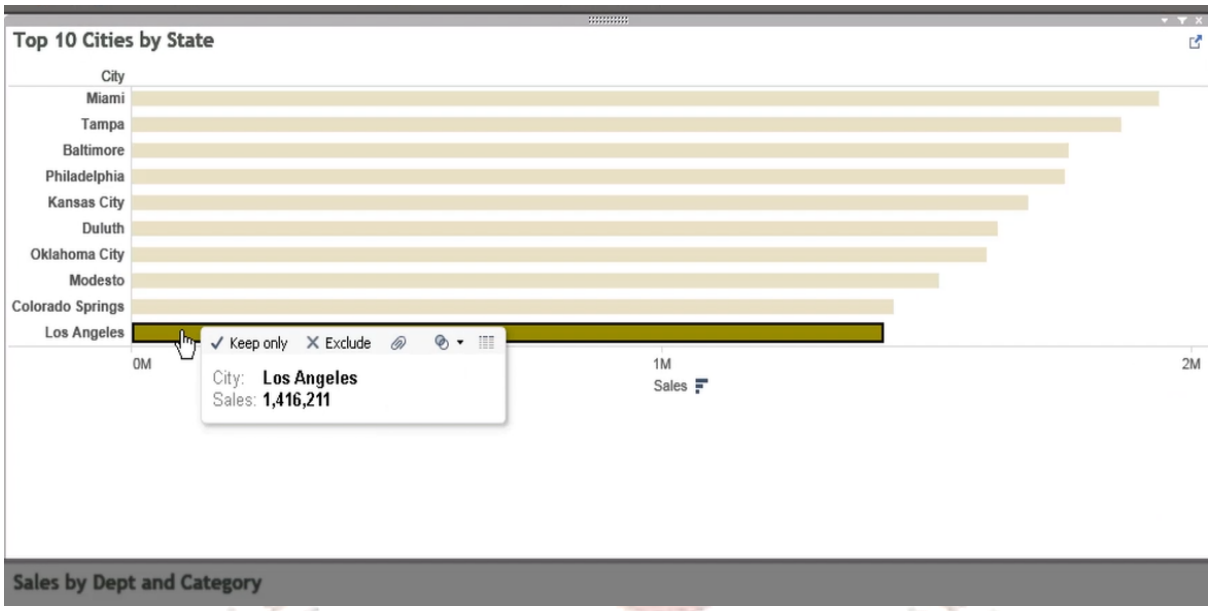


Fig 11.15 Top ten cities by state

Suppose you want to have one top view and two bottom views. When this is the case, place a vertical container at the top of your dashboard layout. Add a horizontal container at the bottom. Drop the view to see at the top of the dashboard into the vertical container, and the two views to see at the bottom into the horizontal container. This is now the layout for the dashboard. So, depending on the views to be placed and where they need to be placed, it is necessary to decide whether horizontal or vertical containers is required.

Step 2: The layout containers always have a blue border when highlighted. Layout containers in general are always pushed to the background of view and it's very hard to select. So an easy way to select any particular element on the dashboard or in specific the layout container would be to click on the element in the layout section, especially when there are multiple or nested container objects.

An easiest way to select layout container is illustrated below. Initially, select the gray container. Then use the dropdown arrow at the top right corner and choose Select Layout Container. Now this highlights the layout container within which the particular view has been placed. Secondly, add a web page to dashboard. we have an option that allows to add a web page or a live connection. This option lets to embed any web page directly within the Dashboard, which in fact is a live connection that displays the details of the page whenever the dashboard is open. Suppose there is a finance dashboard and the interest is in seeing the

daily stock market status within the Dashboard. This could be done by adding the web page with the URL pointing to that particular web page.

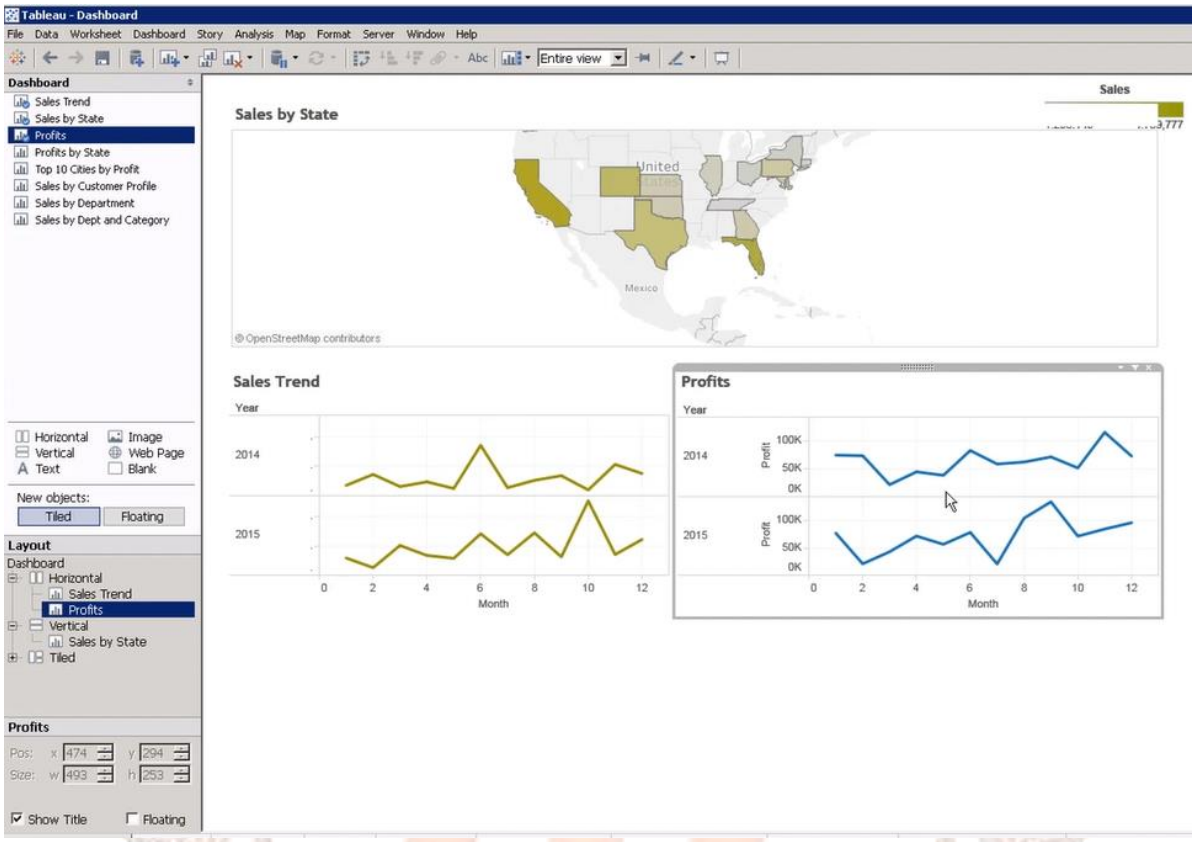


Fig 11.16 Formatting Dashboard

Step 3: Initially, add Horizontal container to the bottom. Track the web page object onto this horizontal container. Enter the webpage address for the Yahoo finance page and click OK. This will bring in the details of the current stock market status from the Yahoo web page and populate it within this object container. To add an image, we simply click on the image object and drag it to the location where the image has to be added. Go ahead and add a logo for this dashboard. Then, pick the PNG image or the logo image file that has been already saved on the desktop. Once it's added, it is possible to center it or fit it by choosing an option from the menu. By right clicking on the image from the menu, it is possible to add or set a URL to make this image linkable to a webpage.

Step 4: Go ahead and create the sample dashboard for this session using our views that have already been created on.

Step 5: Start with adding these views. Sales Strength, Sales by State and the top ten cities by state. Go ahead and add a horizontal container for the dashboard title.

Step 6: Go ahead and drop a text container to add the dashboard title. Drag one more horizontal container to a dashboard layout space.

Step 7: Go ahead and arrange all these three views that have been dropped onto our workspace into this container. Expand the size so that all three views fit into the first row within this container.

Step 8: Adjust the size by clicking on the arrow mark on the horizontal container and also the views to set the size. To format the dashboard. Click on dashboard and choose Format.

Step 9: Change the default Dashboard shading to another color if preferred. Change the dashboard title, font, alignment, shading and so on within this format. Dashboard options. Go and format the layout containers. Finally, change the color to a medium gray.

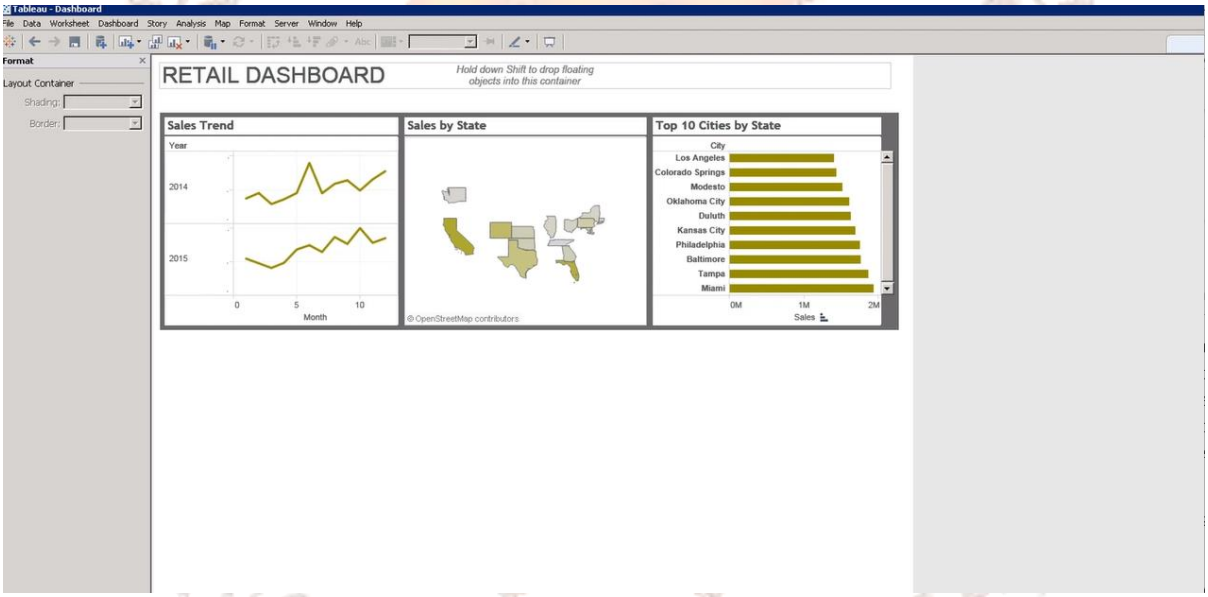


Fig 11.17 Sample Dashboard

Step 10: To change the view colors and format them, it is necessary to directly do it in within the sheet view. This could be done by going back to the view or by right clicking with a view within the dashboard and choosing format.

SELF-ASSESSMENT QUESTIONS – 2

3. The distribution of a single continuous measure can be visualized using _____plot and _____.

5. ADDING INTERACTIVITY TO DASHBOARDS

The process of adding interactivity give the user more power, allowing them to make the analysis much easier. Dashboard Actions are the elements within tableau that helps to have context and provide interactivity to your data in your Dashboard. Various actions are available in a dashboard. Click on dashboard and choose actions. First we have the highlight actions. Highlight Actions allows to call attention to specific points of interest by highlighting or coloring certain marks and dimming the others. Easiest way to do highlighting is by manually selecting a group of marks or a particular mark to highlight.

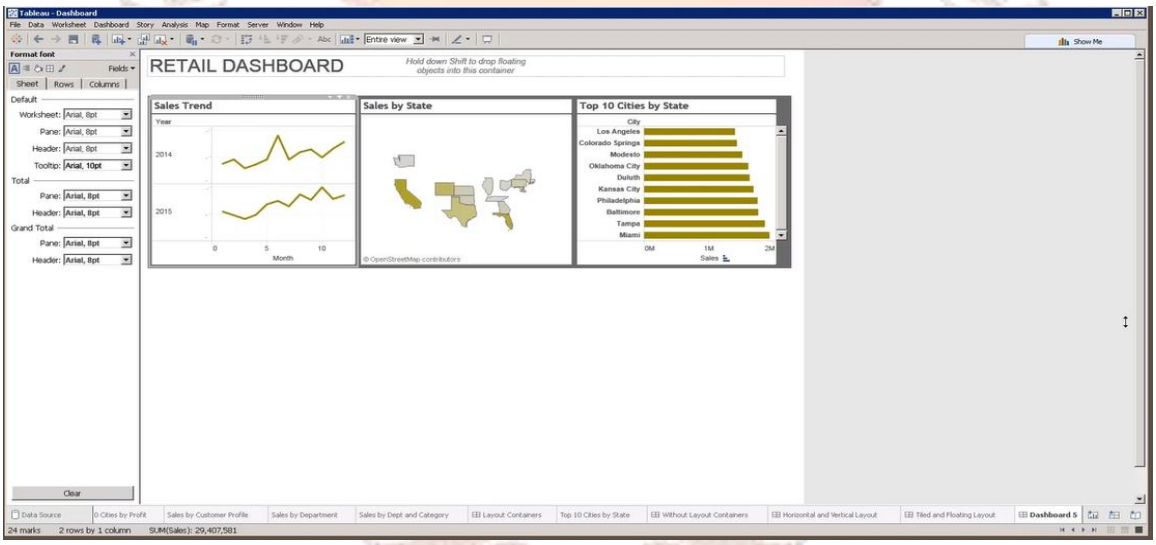


Fig 11.18 Interactivity to dashboards

In the Fig 11.5.1 the mail and therefore only the male category is being highlighted in all the other views. In this case, the mail has been highlighted, but it is possible to highlight a single or even multiple items within Legend. This option can either be enabled or disabled from within the toolbar option at the top. The steps to add interactivity dashboard is illustrated below

Step 1: The option to highlight based on criteria or custom criteria has to be defined. This is done through advanced highlight actions. On a worksheet, go ahead and click on Worksheet Actions. In the Actions dialog box, click the Add Action button and select Highlight. Add a name here and then select the trigger option. To trigger the action on Hover, select on Menu. Hover is useful especially in a Dashboard when the mouse over a mark in the view and the corresponding marks in all other views are highlighted. Or rather the particular action that has been added is run.

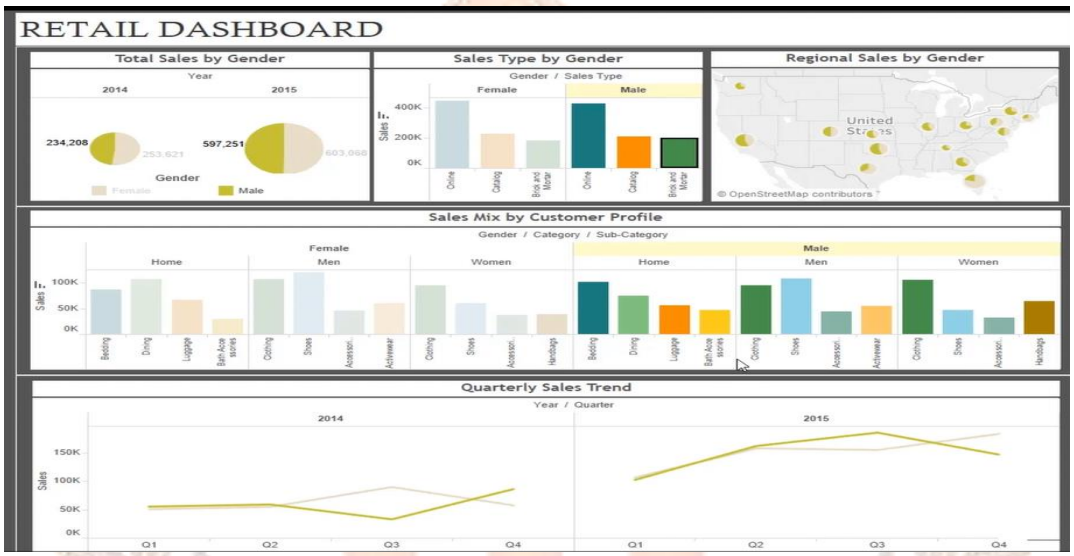


Fig 11.19 Highlight actions

Step 2: Select the highlight marks based on a click action. Click on a mark, the action is run. Finally, is the Menu option. This allows us to highlight by selecting an option from the context menu by right clicking on a mark. Go ahead and select the target sheets.

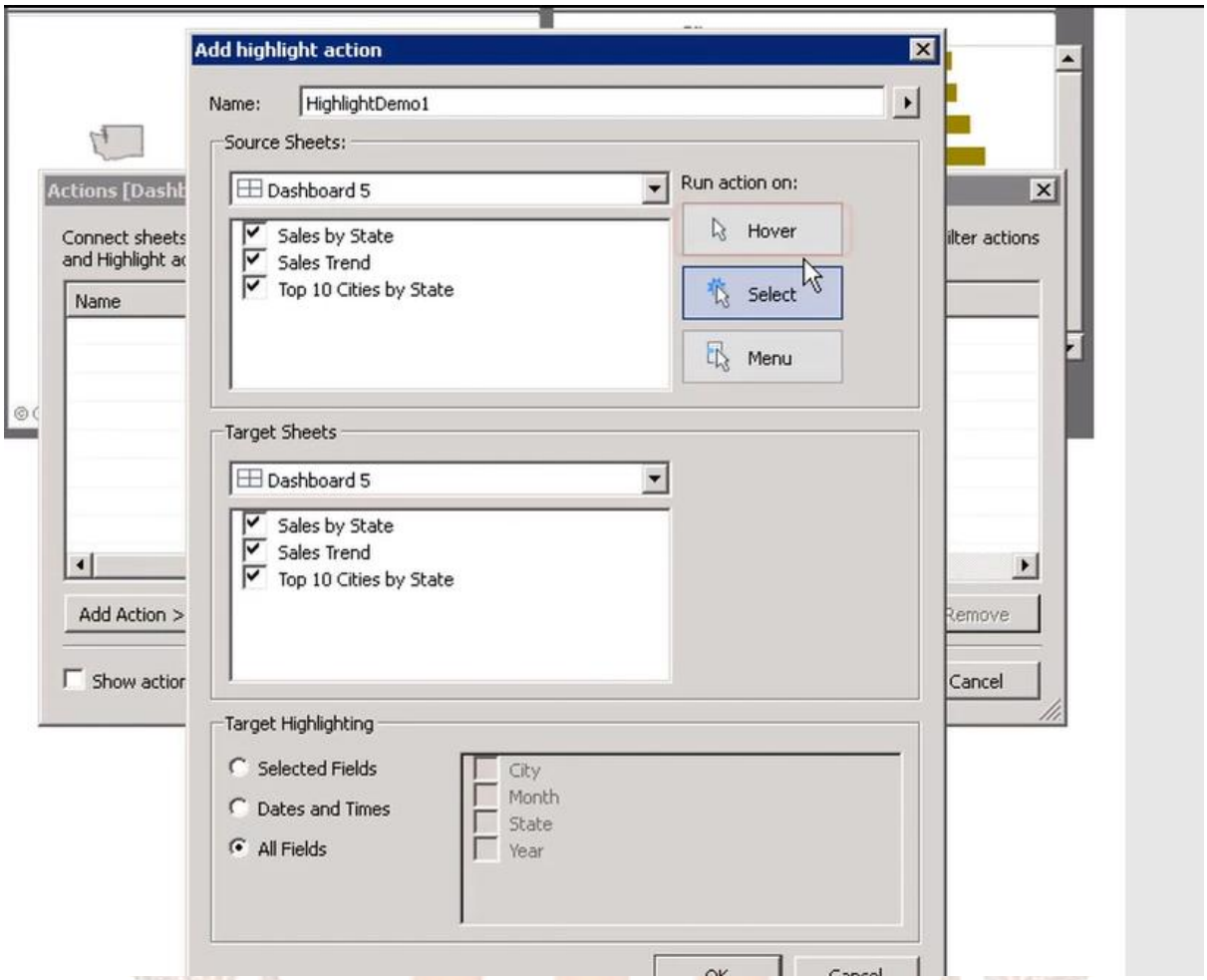


Fig 11.20 Adding Highlight actions

Step 3: Select the source sheet as Sales by State year and select the target sheets as Sales by State, sales Trend and Top Sensitives by State.

Step 4: Go ahead and complete rest of the Dashboard offline so that it is possible to review it further. Next, check if the layout is suitable for the end presentation or device, and then we can specify the overall size in the Dashboard menu and if it needs to be automatic, or if it automatically needs to resize to fit the window it is displayed in.

Step 5: Check if any unwanted items are present so that they can be cleaned up.

Step 6: Shading affects the dashboard objects itself and not the view themselves.

6. BUILDING STORIES

A story is a sequence of worksheets or dashboards that convey information in a narrative manner so it is possible to indicate the way plot is connected and provide some context and make a company spelling case using a storyline so that the business decision making process is much easier. Each individual sheet in a tableau story is called a story point and it's not a static presentation .It is a live connection to data and can change as the underlying data changes. Stories work great as a presentation medium, making the narrative flow easier to present to the audience. Once a click is done on the new story window, a new layout window opens up with the left panel listing the available worksheets and the dashboards that can be added to the story.

6.1 Steps to Create Story Using Tableau

Step 1: Choose to display or hide the Navigator buttons and specify the size and choose to show or hide the title. Go ahead and choose a size from the predefined sizes

Step 2: Go ahead and choose a desktop size. To edit the title, double click and choose your report title.

Step 3: Add the sheet that wanted to be displayed as the first story point. Go ahead and add sales trend. It is also possible to customize these sheets within the story point by selecting a particular range of marks or by sorting them in a particular order, or by filtering it to a particular field and retaining them as part of the story. Any customization that is done on the Story Point sheet will not be automatically updated in the original sheet. However, edits in the original sheet, will be carried forward to the story points.

Step 4: In order to format the story pane, click on Story and choose Format . It is possible to change the shading, or we can even set a color and set a transparency level, or set a Navigator shading, or set the alignment, or the font and so on. To add a caption to each one of the story points, double click and enter the story point name.

Step 5: Create a new story point, click on the new blank point. And now drag another sheet onto this sheet and drag another sheet onto your story point.

Step 6: To delete the story point, click on the delete icon that appears right next to the story point. To rearrange or remove story points, just drag a sheet and drop it in any order is required. In order to duplicate story points click on the duplicate icon. This will create a copy of the story point where a separate caption is created. To navigate between the story points or use the back forward button in case there are a lot of story points.

Step 7: If the navigator buttons are not required, choose deselect. There are different tabs for the different story points. Once the story is created view the story or rather present using the presentation mode on the top. This will give a magnified view of the story.

SELF-ASSESSMENT QUESTIONS – 3

4. Can a URL activity on a dashboard question be conveyed to an open a Site page inside a dashboard instead of opening the framework's internet browser? Yes/No

Answers

1. Reporting requirements
2. Dashboard
3. Box plot and histogram
4. Yes, with the utilization of a Site page protest
5. Sum
6. Countd

TERMINAL QUESTIONS

1. Give a brief about the tableau dashboard?

Tableau dashboard is a group of various views which allows you to compare different types of data simultaneously. Datasheets and dashboards are connected if any modification happens to the data that directly reflects in dashboards. It is the most efficient approach to visualize the data and analyze it.

2. Define Page Shelf in Tableau?

Page shelf breaks the views into a series of pages. It displays an alternate view on each page. Due to this feature, you can analyze the effect of each field on the rest of the data in the view.

3. Define the story in Tableau?

The story can be defined as a sheet which is a collection of series of worksheets and dashboards used to convey the insights of data. A story can be used to show the connection between facts and outcomes that impacts the decision-making process. A story can be published on the web or can be presented to the audience.

4.Give an overview of the fact and dimensions of the table?

Facts are numeric measures of data. They are stored in fact tables. Fact tables store that type of data that will be analyzed by dimension tables. Fact tables have foreign keys associating with dimension tables.

Dimensions are descriptive attributes of data. Those will be stored in the dimensions table. For example, customer's information like name, number, and email will be stored in the dimension table.

5. State some ways to improve the performance of Tableau
   - Use an Extract to make workbooks run faster
   - Reduce the scope of data to decrease the volume of data
   - Reduce the number of marks on the view to avoid information overload
   - Try to use integers or Booleans in calculations as they are much faster than strings
   - Hide unused fields
   - Use Context filters
   - Reduce filter usage and use some alternative way to achieve the same result
   - Use indexing in tables and use the same fields for filtering
   - Remove unnecessary calculations and sheets
6. Explain different connection types in Tableau?

There are 2 connection types available in Tableau.

Extract: Extract is a snapshot of data that will be extracted from the data source and put into the Tableau repository. This snapshot can be refreshed periodically fully or incrementally. This can be scheduled in Tableau Server.

Live: It creates a direct connection to the data source and data will be fetched directly from tables. So, data will be up to date and consistent. But, this also affects access speed.

7. Explain how many types of filters are available in Tableau?

Filters are used to provide the correct information to viewers after removing unnecessary data. There are various types of filters available in Tableau.

- **Extract Filters –** Extract filters are used to apply filters on extracted data from the data source. For this filter, data is extracted from the data source and placed into the Tableau data repository.
- **Datasource Filters –** Datasource filters are the same as extract filters. They also work on the extracted dataset. But, the only difference is it works with both live and extract connections.
- **Context Filters –** Context Filters are applied on the data rows before any other filters. They are limited to views, but they can be applied on selected sheets. They define Aggregation and Disaggregation of data in Tableau
- **Dimension Filters –** Dimension filters are used to apply filters on dimensions in worksheets. Dimension filters are applied through the top or bottom conditions, formula, and wildcard match.
- **Measure Filters –** Measure filters are applied to the values present in the measures.

8. Differentiate between Tiled and Floating in dashboards?

In a tiled layout, items don't overlap. The layout will be adjusted according to dashboard size. In the floating layout, items can be placed on some other layers. Floating items can have fixed positions and sizes.

9. State the components of the dashboard?

The dashboard consists of 5 components.

- **Web:** it consists of a web page embedded in the dashboard.
- **Horizontal component:** it is a horizontal layout container in which we can add objects.
- **Vertical component:** it is a vertical layout container in which we can add objects.
- **Image Extract:** it allows you to upload an image to the dashboard from a computer.
- **Text:** it is a small Wordpad where we can format and edit the text.

8. SUMMARY

Let us recapitulate the important concepts discussed in this unit:

- To use the dashboards, interactive dashboards and formatting dashboards
- Provides explanations for when and why you should use a dashboard and story to visualize your data.
- Provides explanation about steps involved in creating a dashboard.
- Provides explanation about steps involved in creating a story.

7. GLOSSARY

Let us have an overview of the important terms mentioned in the unit:

Dashboard: A dashboard is a collection of several views, letting you compare a variety of data simultaneously.

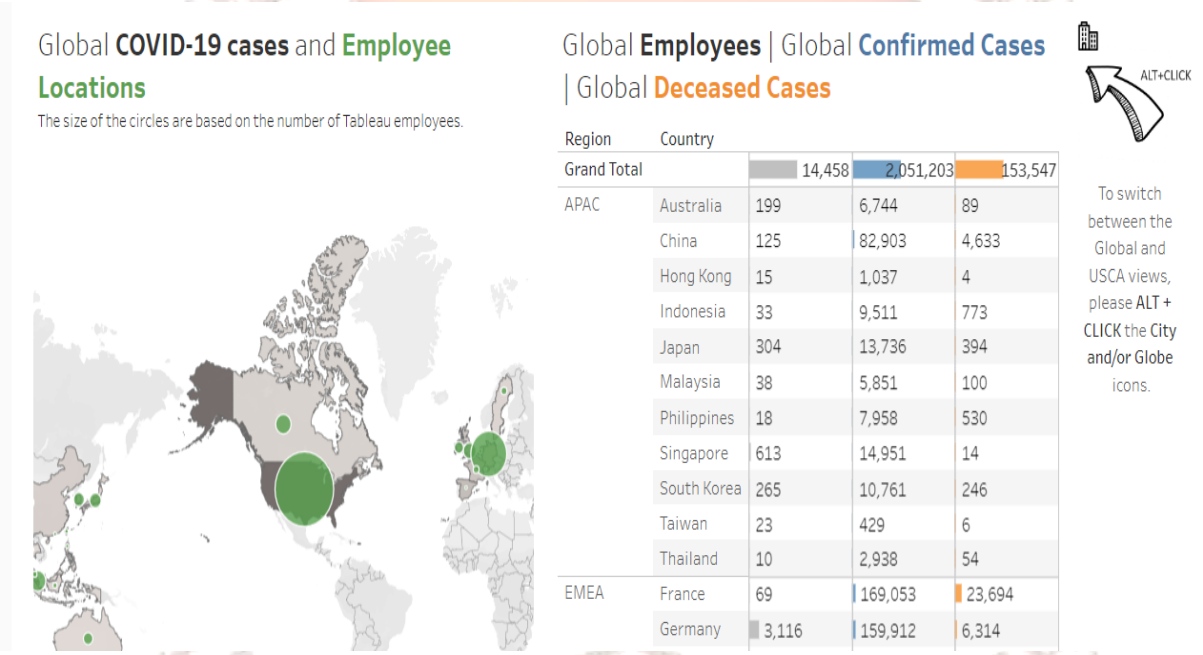
Story: In Tableau, a story is a sequence of visualizations that work together to convey information.

8. CASELET

1. Finance: Understanding and sharing variances

Given the rapid pace of changes in the economy and business operations, it can be harder than ever to manage budgets, expenses and forecasts. The finance team's dashboards provide up-to-date results and enable the team to change forecasts as uncertainty decreases. By sharing this dashboard with leaders across the business, the finance team supports strategic decision-making and resource allocation.

The COVID-19 pandemic has forced human resources departments to ask new questions around employee safety and working from home. The Tableau people analytics team used data to inform their response: They enriched their data with publicly available information and by gathering new data from an employee survey.



Region	Country	Employees	Confirmed Cases	Deceased Cases
Grand Total		14,458	2,051,203	153,547
APAC	Australia	199	6,744	89
	China	125	82,903	4,633
	Hong Kong	15	1,037	4
	Indonesia	33	9,511	773
	Japan	304	13,736	394
	Malaysia	38	5,851	100
	Philippines	18	7,958	530
	Singapore	613	14,951	14
	South Korea	265	10,761	246
	Taiwan	23	429	6
	Thailand	10	2,938	54
EMEA	France	69	169,053	23,694
	Germany	3,116	159,912	6,314

References and Suggested Reading

1. Joshua N. Milligan, Learning Tableau 2022: Create effective data visualizations, build interactive visual analytics, and improve your data storytelling capabilities, 5th Edition 5th ed. Edition
2. Visual Analytics with Tableau, Paperback, 31 May 2019.

9. CONCEPTUAL MAP

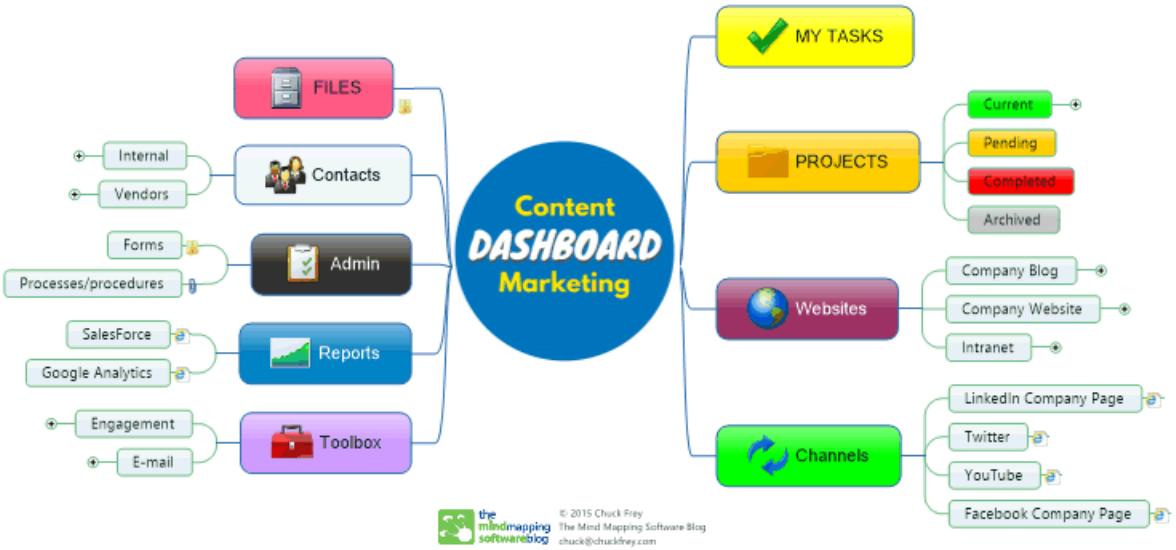


Fig 11.21